

Case Diagnosis

A 73-year-old man with acute heart failure (EF <14%) from non-ischemic cardiomyopathy, valvular disease, and atrial fibrillation/flutter underwent HeartMate 3 left ventricular assist device (LVAD) implantation, followed by inpatient cardiac rehabilitation.

Case Description

His peri-LVAD course included a one-month ICU stay requiring hemodynamic support with Impella bridging to HeartMate 3 implantation and aortic valve replacement. Two months into acute care recovery, he developed critical illness myopathy with severe deconditioning, cognitive decline, dysphagia, and sacroccocygeal ulcers, prompting admission to the acute rehabilitation unit (ARU) for advanced cardiac rehabilitation.

Assessment & Results

Over two weeks in the ARU, he received ≥3 hours/day of PT, OT, and speech pathology alongside 24/7 rehabilitation nursing and daily physiatry oversight. Care integrated LVAD monitoring, device parameter optimization, and anticoagulation management with warfarin. Caregivers completed structured LVAD education and safety training prior to discharge home with home health services and planned outpatient cardiac rehabilitation.

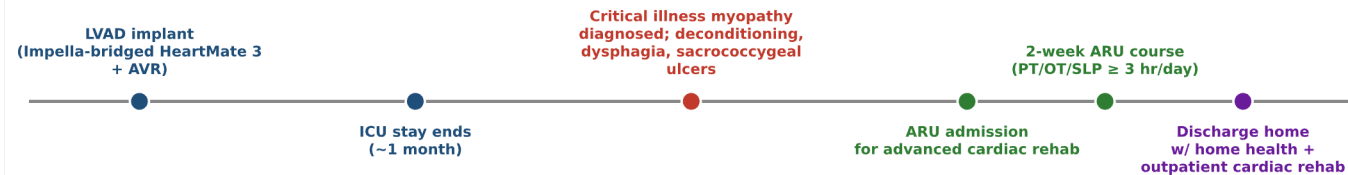
Discussion

Although LVAD technology has transformed survival in end-stage heart failure, rehabilitation pathways after implantation remain underreported. LVAD-ready ARUs require specialized infrastructure, multidisciplinary expertise, and coordinated workflows integrating physiatrists, cardiologists, therapists, and specialized nursing. Device-specific education and emergency protocols are essential for patient safety.

Functional Status: ARU Admission vs. Discharge

Functional Domain	ARU Admission	ARU Discharge
Hygiene & grooming	Minimal assist	Set-up assist
Bed mobility	Minimal assist	Set-up assist
Transfers	Minimal assist	Set-up assist
Sit-to-stand	Minimal assist	Set-up assist
LVAD self-management	Moderate assist	Min / supervised standby
Toileting	Moderate assist	Min / supervised standby
Ambulation (FWW)	10 ft, moderate assist	150 ft, supervision
Cognition	Impaired	Returned to baseline
Dysphagia	Present	Resolved

Hospital Course: LVAD Implantation to Discharge



Conclusion

LVAD-capable ARUs demonstrate how specialized inpatient rehabilitation can translate technological survival advances into meaningful functional recovery, highlighting the expanding role of physiatry in the continuum of care for complex patients.

